

The empirical recurrence for OEIS A230187, recovered from its tail

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8 September 2026

Abstract

OEIS A230187 records “Empirical recurrence of order 93” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 176$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 97$. Its last coefficient is $c_{93} = 100576849920 \neq 0$, so no further tail satisfies anything shorter; any order-93 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A230187 is “Number of $4 \times n$ 0..2 arrays $x(i,j)$ with each element horizontally or antidiagonally next to at least one element with value $2 - x(i,j)$.”. It has offset 1 and begins

0, 129, 1971, 73857, 2432241, 82359177, 2770525761, 93340154655, ...

The entry states:

Empirical recurrence of order 93 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 08:42:43, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 176$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 176$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 93: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 93.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 352$ exact terms from $n = 4$ on, it returns order 93 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{93} c_i a(n-i),$$

c_1	24	c_{25}	−359764711698039	c_{49}	34353579251620344	c_{73}	−6595149519375801
c_2	366	c_{26}	68173923728738	c_{50}	64416448136351774	c_{74}	−10905632268302039
c_3	−638	c_{27}	−256348832877991	c_{51}	−44126801961806058	c_{75}	−10235813955343556
c_4	−23209	c_{28}	1026368405776089	c_{52}	−37330493474341396	c_{76}	4273227292002252
c_5	−40841	c_{29}	401574205971960	c_{53}	−90196316083133896	c_{77}	6025966207599986
c_6	653432	c_{30}	550292942680128	c_{54}	−93505254924692927	c_{78}	985362273645081
c_7	2684226	c_{31}	−356730490872096	c_{55}	59125472828128746	c_{79}	2959061580886021
c_8	−8543293	c_{32}	−1122927441525696	c_{56}	−90115107934008633	c_{80}	3131706796390161
c_9	−69180034	c_{33}	−2010153210794421	c_{57}	190663650665842543	c_{81}	136680825414901
c_{10}	−4579304	c_{34}	−2792799202597295	c_{58}	−84255694888596750	c_{82}	−1595907797415238
c_{11}	964195951	c_{35}	−2498832434663721	c_{59}	204835026838167232	c_{83}	−684694577444976
c_{12}	2007236888	c_{36}	394047187253230	c_{60}	−100506050829456050	c_{84}	−84725798677020
c_{13}	−6441096155	c_{37}	4829950128431460	c_{61}	70065721359059903	c_{85}	−155135142012996
c_{14}	−33609712346	c_{38}	9962776724650073	c_{62}	−128748934626948775	c_{86}	−61366151006336
c_{15}	−10200678360	c_{39}	10147684873977470	c_{63}	−27989086473399141	c_{87}	24460124169664
c_{16}	257696845221	c_{40}	7853505515809464	c_{64}	−27193689148297114	c_{88}	−1947874295888
c_{17}	607454468189	c_{41}	3007316611524083	c_{65}	19980846306628209	c_{89}	16579106067616
c_{18}	−665887026197	c_{42}	−22274247371680945	c_{66}	14562982190124527	c_{90}	16755294986560
c_{19}	−4869776695664	c_{43}	−14386013644609604	c_{67}	194521548857833	c_{91}	2869180890368
c_{20}	−5055712089869	c_{44}	−49850207255666220	c_{68}	43203152716460734	c_{92}	2337236202496
c_{21}	14917392970463	c_{45}	7097641270624030	c_{69}	29247544349833199	c_{93}	100576849920
c_{22}	39955863388252	c_{46}	−12562071233335462	c_{70}	170027294986271		
c_{23}	13833923020836	c_{47}	45456658581137994	c_{71}	−26519012777864130		
c_{24}	−9504850163033	c_{48}	51145955586245851	c_{72}	−2166128486339589		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 97$, and it is the recurrence OEIS A230187 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{93} - c_1x^{93-1} - \dots - c_{93}$. Its constant term is $-c_{93} = -100576849920$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 93 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 93, so $\deg q = 93$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 93, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 93, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 100576849920.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-93 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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